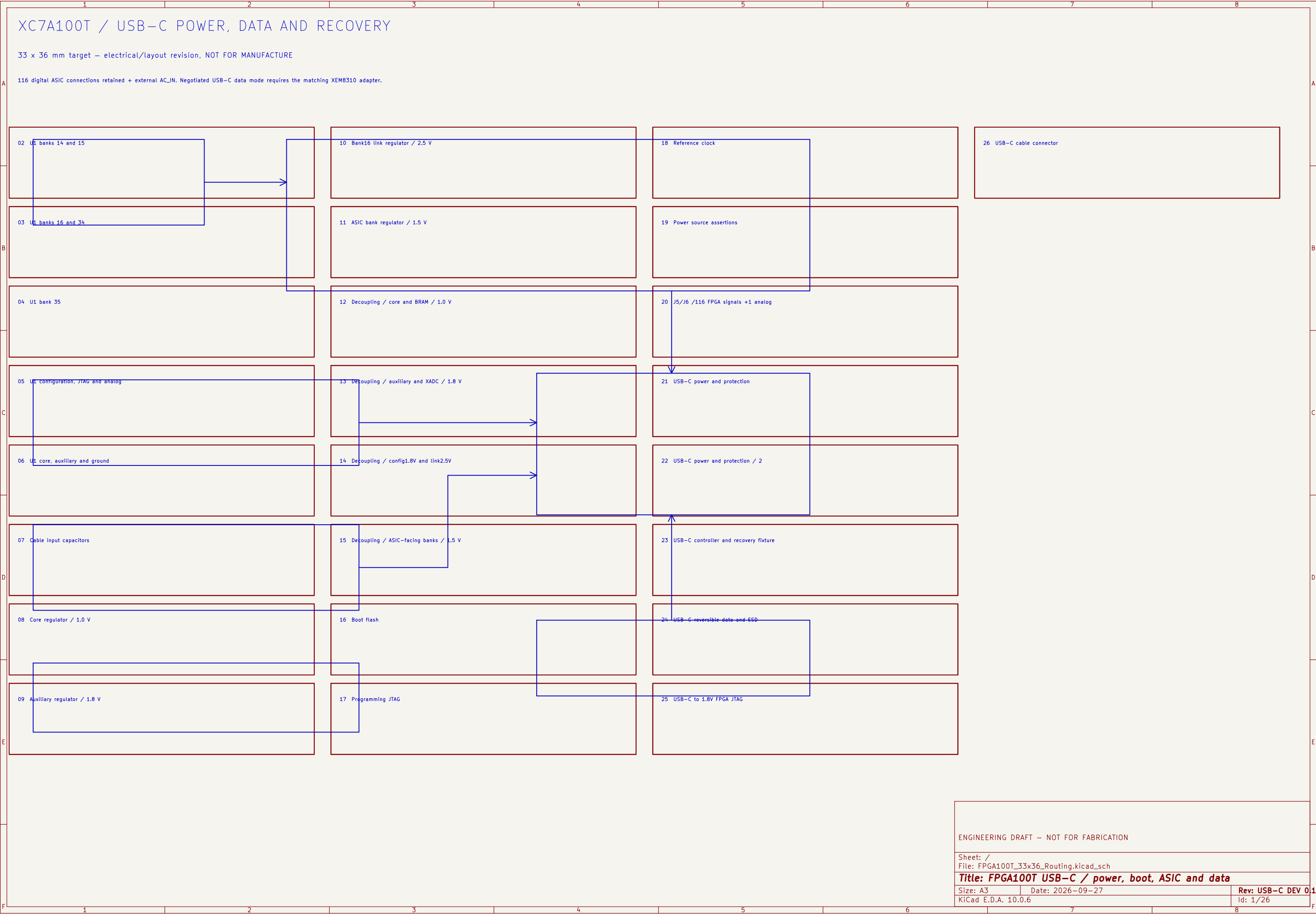


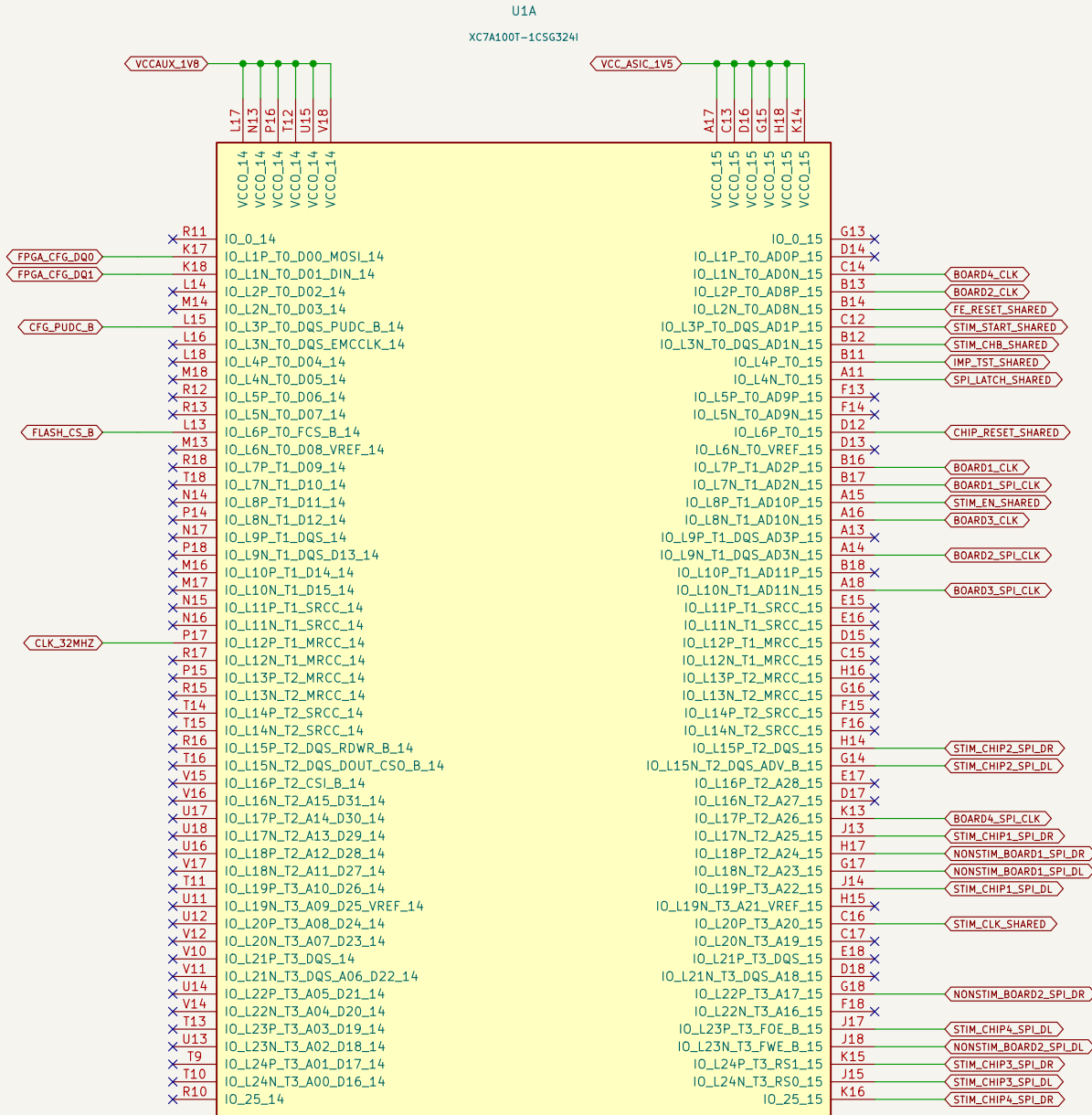


U1 / XC7A100T-CSG324 / I/O banks 14 and 15

116 FPGA-linked signals +1 analog contact. Bank15/34/35 digital limits and clock/timing remain under review.

AC_IN is analog0–1.5 V: J5.46 only, FPGA A13 NC. IMP_TST: assumed0/1.5V digital test control; polarity and inactive state still pending.

A13 intentionally NC. AC_IN is analog0–1.5 V and is not an FPGA GPIO.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /U1 banks 14 and 15/

File: fpga_unit_01.kicad_sch

Title: U1 / XC7A100T-CSG324 / I/O banks 14 and 15

Size: A3

Date: 2026-09-27

Rev: USB-C DEV 0.1

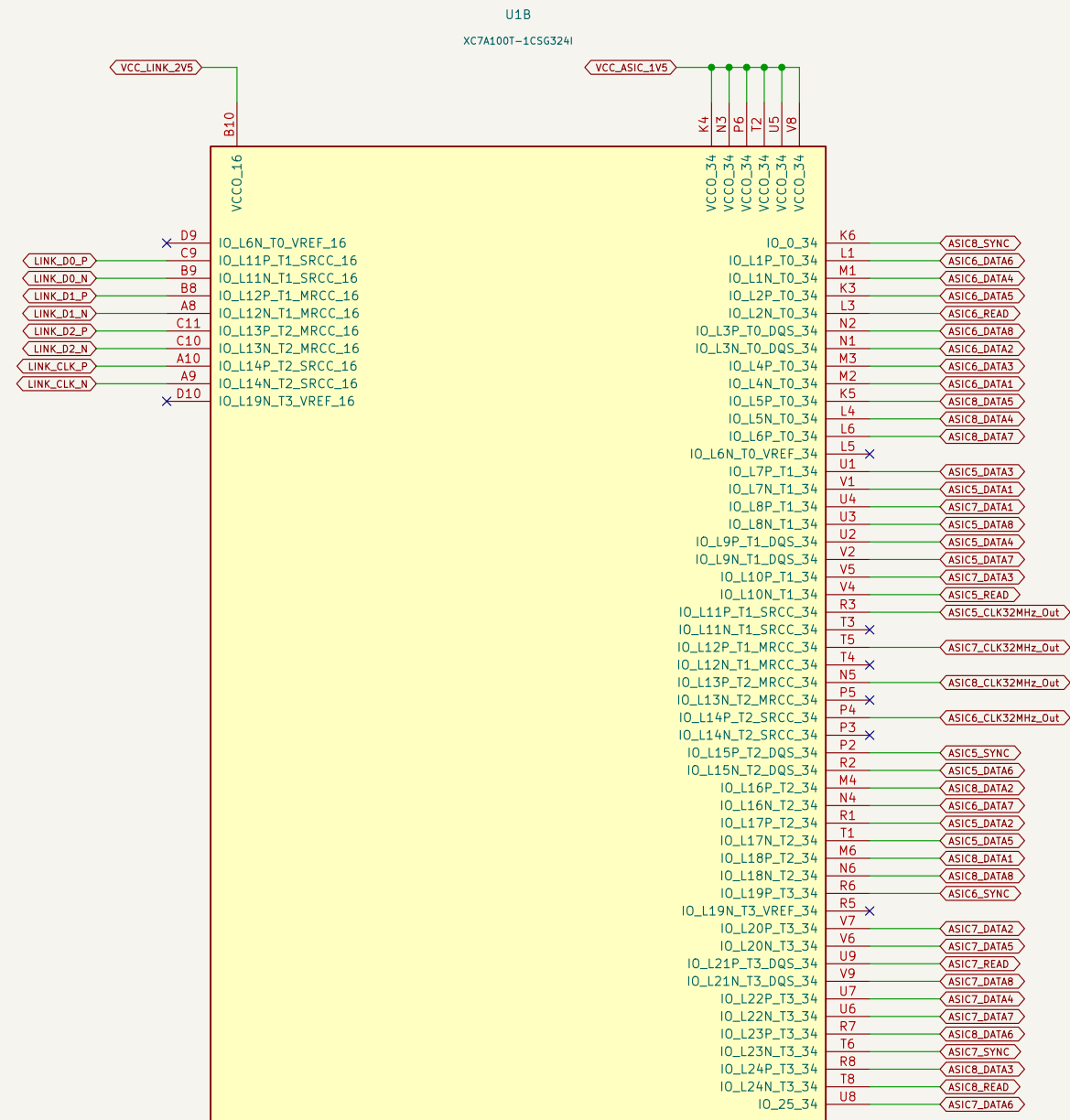
KiCad E.D.A. 10.0.6

Id: 2/26

U1 / XC7A100T-CSG324 / I/O banks 16 and 34

116 FPGA-linked signals +1 analog contact. Bank15/34/35 digital limits and clock/timing remain under review.

AC_IN is analog0-1.5 V: J5.46 only, FPGA A13 NC. IMP_TST: assumed 0/1.5V digital test control; polarity and inactive state still pending.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /U1 banks 16 and 34/

File: fpga_unit_02.kicad_sch

Title: U1 / XC7A100T-CSG324 / I/O banks 16 and 34

Size: A3

Date: 2026-09-27

Rev: USB-C DEV 0.1

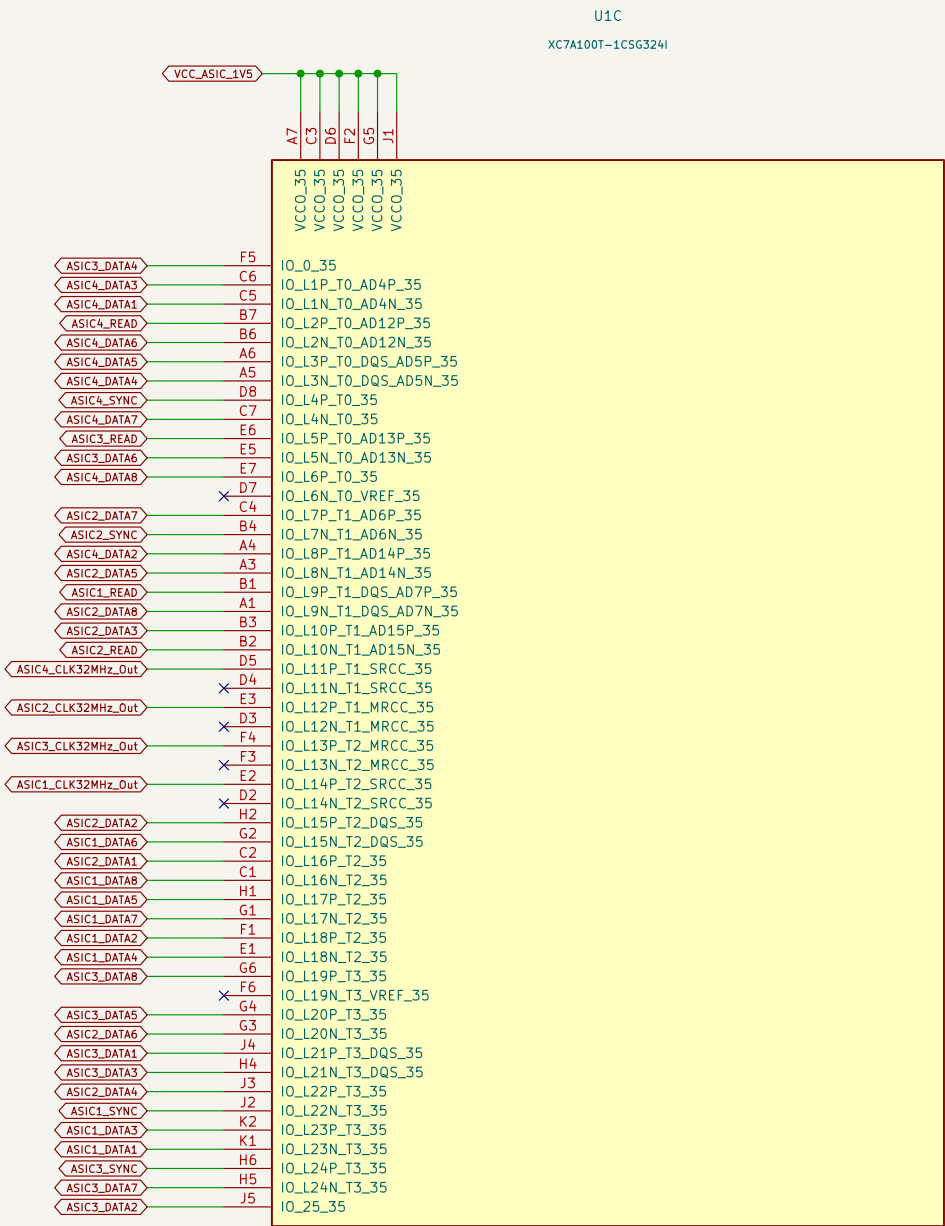
KiCad E.D.A. 10.0.6

Rev. 05D-0
d: 3/26

U1 / XC7A100T-CSG324 / I/O bank 35

116 FPGA-linked signals +1 analog contact. Bank15/34/35 digital limits and clock/timing remain under review.

AC_IN is analog0-1.5 V: J5.46 only, FPGA A13 NC. IMP_TST: assumed0/1.5V digital test control; polarity and inactive state still pending.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /U1 bank 35/

File: fpga_unit_03.kicad_sch

Title: U1 / XC7A100T-CSG324 / I/O bank 35

Size: A3

Date: 2026-09-27

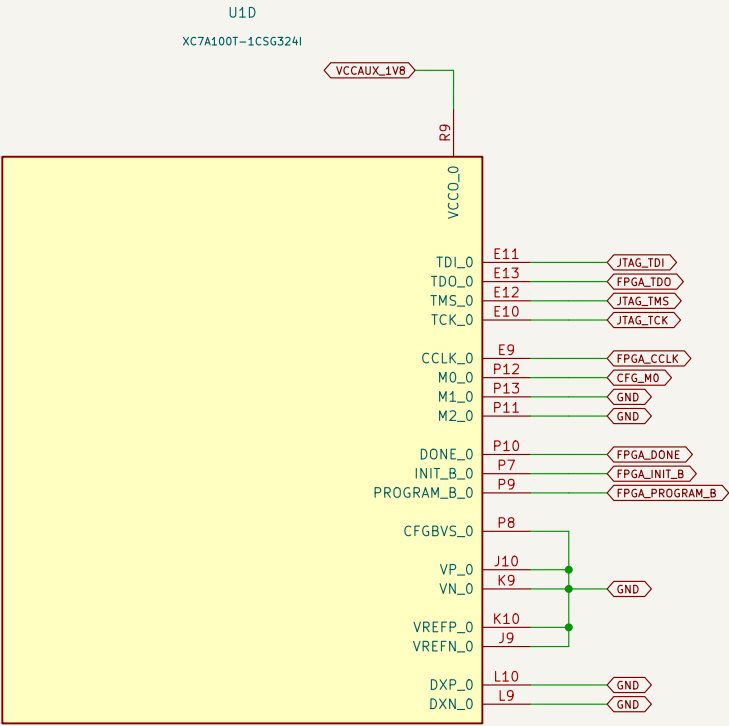
Rev: USB-C DEV 0.1

KiCad E.D.A. 10.0.6

Id: 4/26

U1 / XC7A100T–CSG324 / Configuration, JTAG and analog support

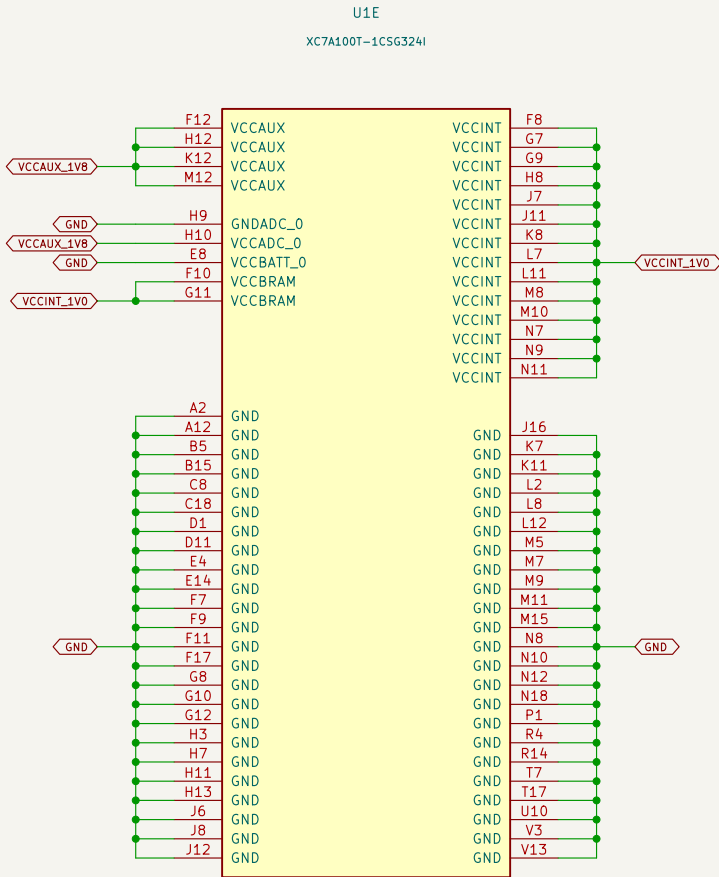
15 x 15 mm CSG324. The117 allocated ASIC contacts comprise116 FPGA–linked signals and1 analog contact. Unassigned FPGA pins are deliberately open; interface details remain under review.



Required–ground correction: unused DXP/DXN grounded per AMD UG475 Table1–12.

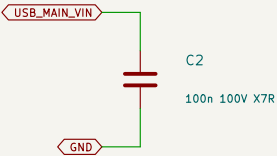
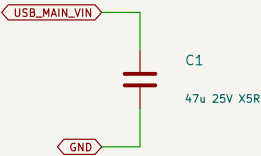
U1 / XC7A100T-CSG324 / Core, auxiliary power and ground

15 x 15 mm CSG324. The 117 allocated ASIC contacts comprise 116 FPGA-linked signals and 1 analog contact. Unassigned FPGA pins are deliberately open; interface details remain under review.



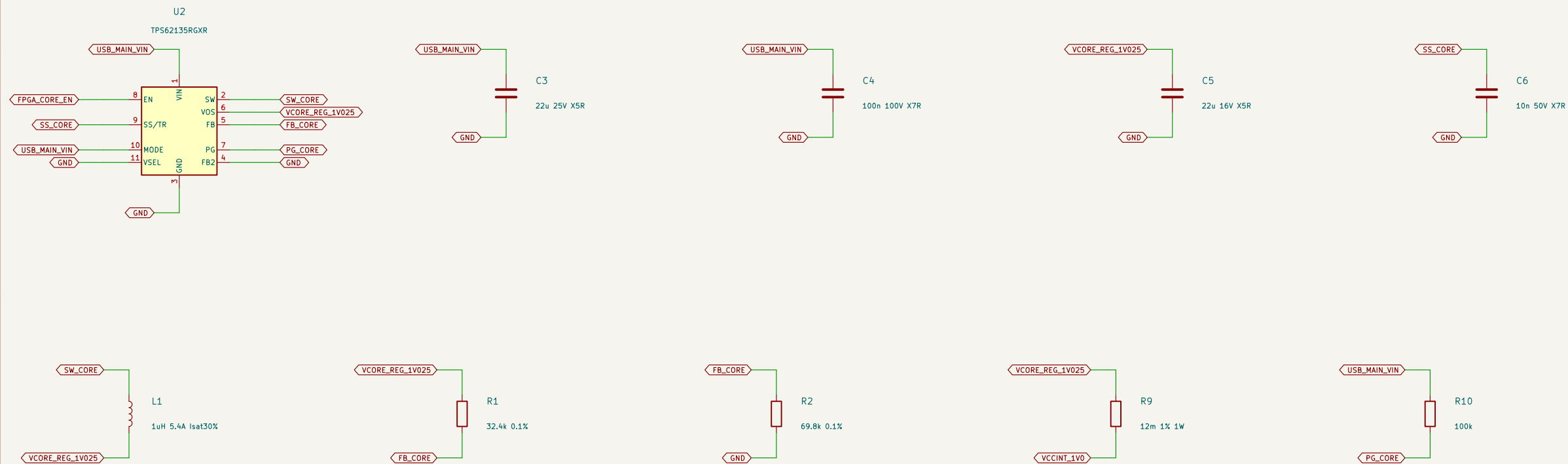
Cable input capacitors

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Core regulator / 1.0 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



FB2 grounded in fixed VSEL-low mode; review inference from internal switch function (TI TPS62135 Fig5).

ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /Core regulator / 1.0 V/

Title: Core regulator / 1.0 V

Size: A3

Date: 2026-09-27

Rev: USB-C DEV 0.1

KiCad E.D.A. 10.0.6

Id: 8/26

Auxiliary regulator / 1.8 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.

A

B

C

D

E

F

A

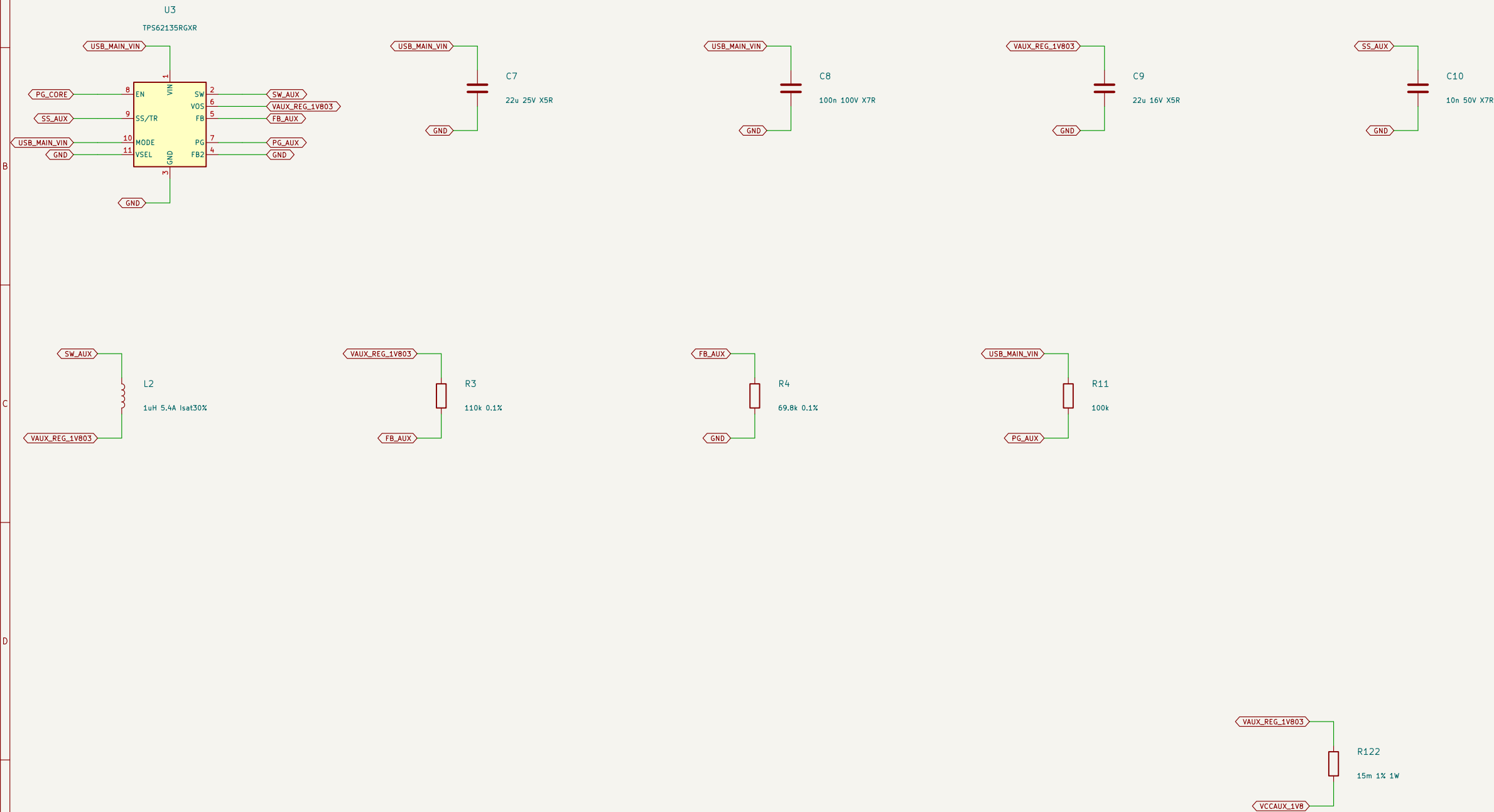
B

C

D

E

F



FB2 grounded in fixed VSEL–low mode; review inference from internal switch function (TI TPS62135 Fig5).

Bank16 link regulator / 2.5 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.

A

B

C

D

E

F

A

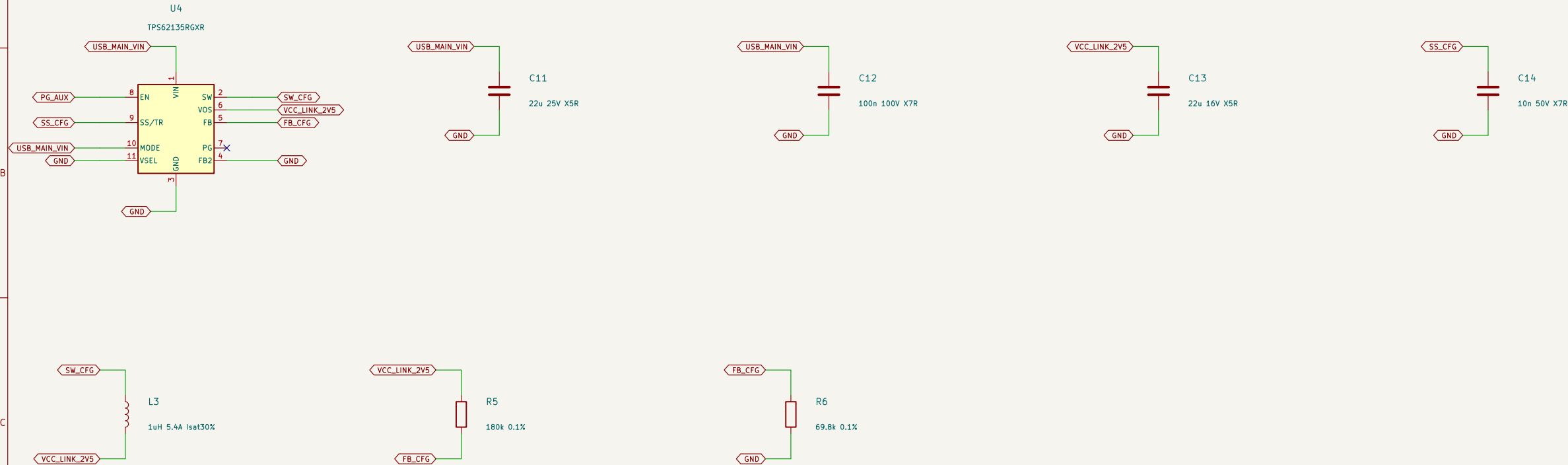
B

C

D

E

F



FB2 grounded in fixed VSEL–low mode; review inference from internal switch function (TI TPS62135 Fig5).

ENGINEERING DRAFT – NOT FOR FABRICATION		
Howard Wang / XC7A100T CSG324		
Sheet: /Bank16 link regulator / 2.5 V/ File: core_04_01.kicad_sch		
Title: Bank16 link regulator / 2.5 V		
Size: A3	Date: 2026–09–27	Rev: USB–C DEV 0.1
KiCad E.D.A. 10.0.6		Id: 10/26

ASIC bank regulator / 1.5 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.

A

B

C

D

E

F

A

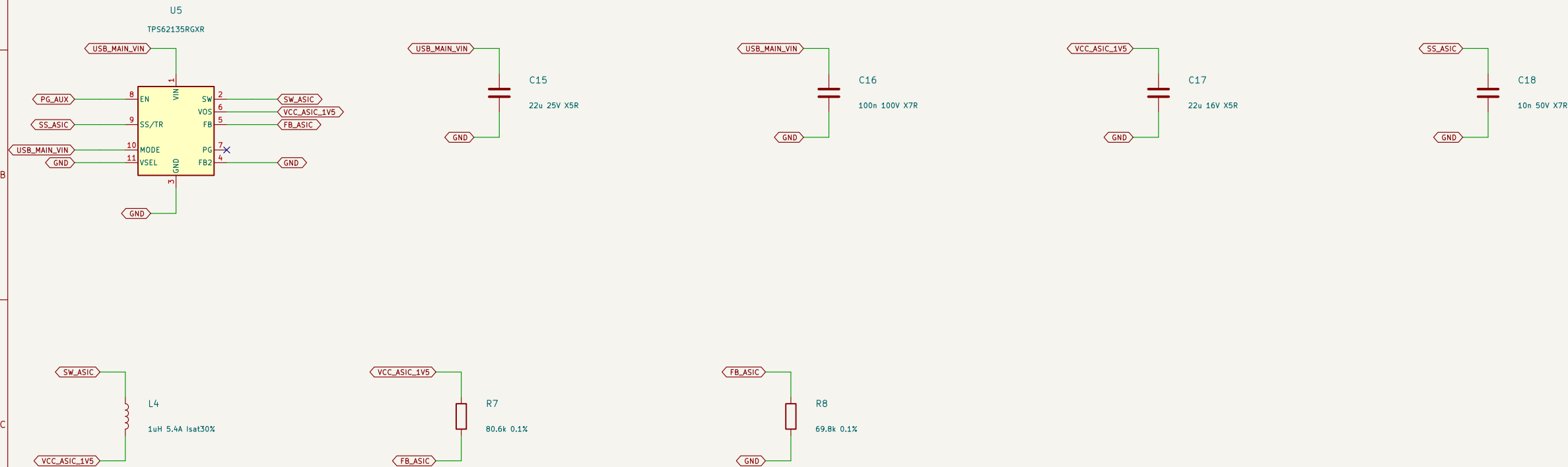
B

C

D

E

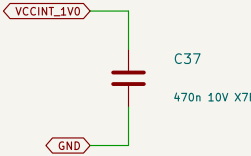
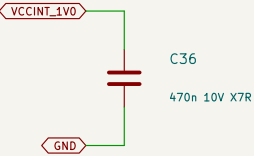
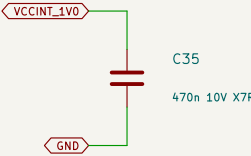
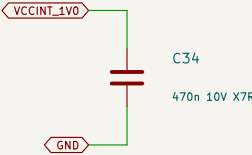
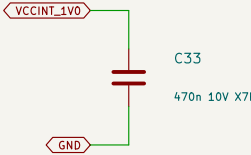
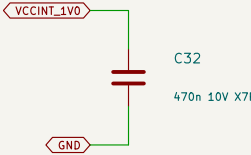
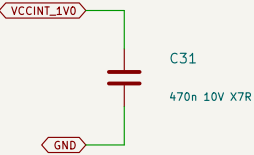
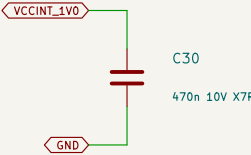
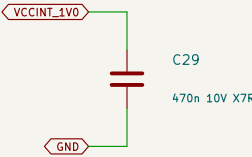
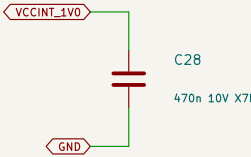
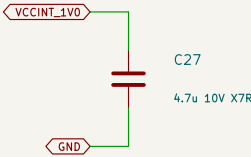
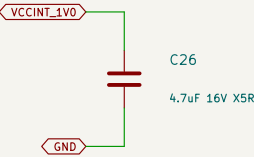
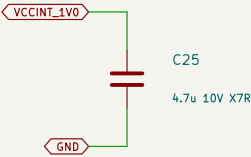
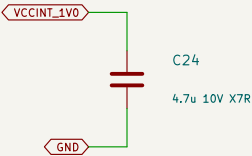
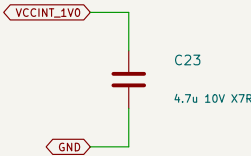
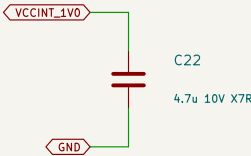
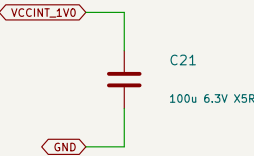
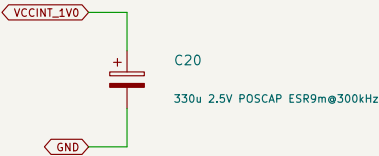
F



FB2 grounded in fixed VSEL–low mode; review inference from internal switch function (TI TPS62135 Fig5).

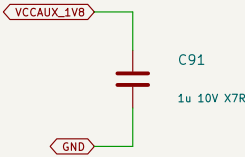
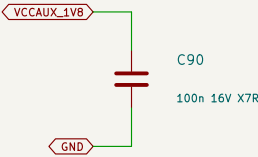
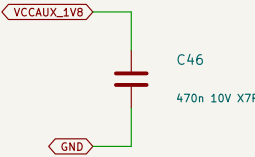
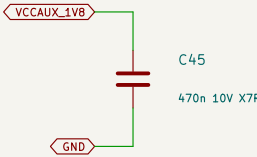
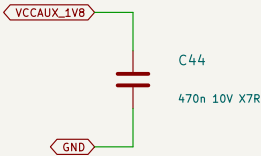
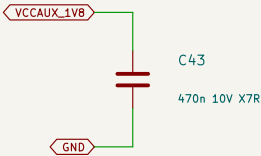
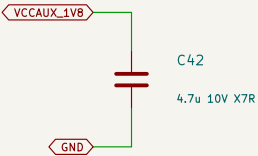
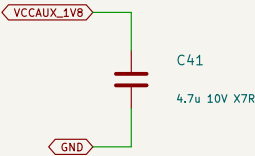
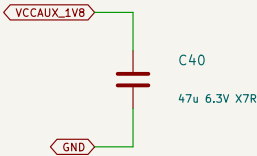
Decoupling / core and BRAM / 1.0 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



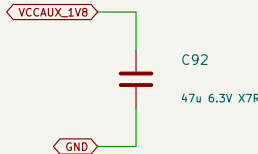
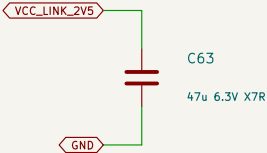
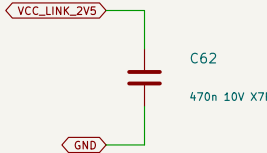
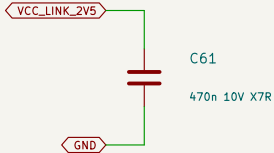
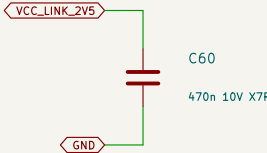
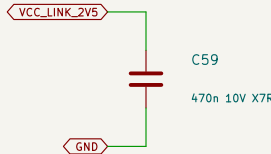
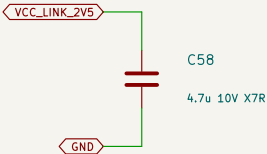
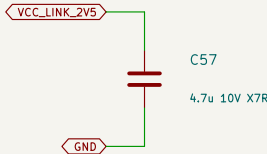
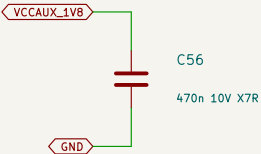
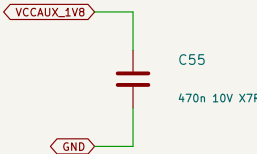
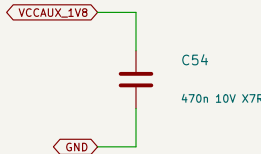
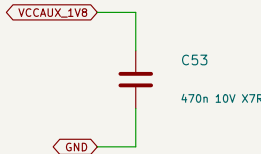
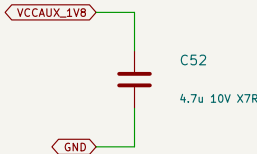
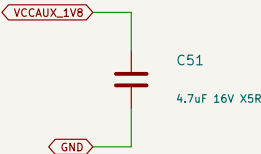
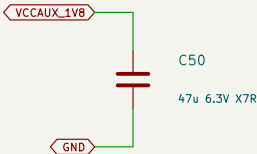
Decoupling / auxiliary and XADC / 1.8 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



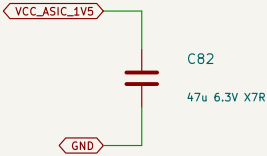
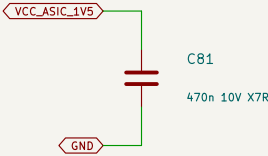
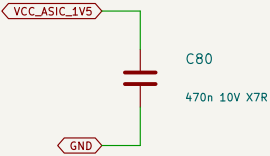
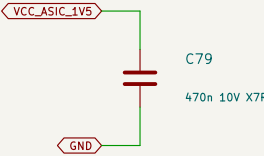
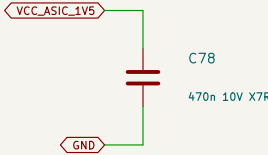
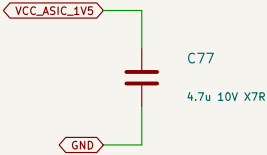
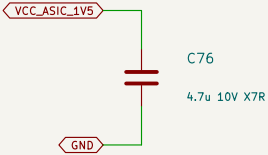
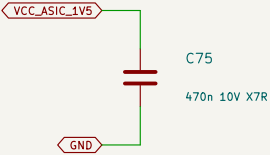
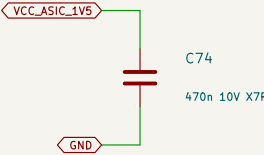
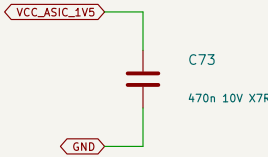
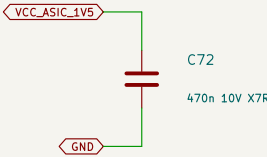
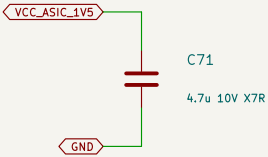
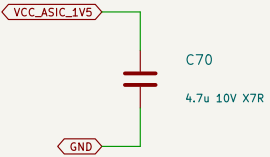
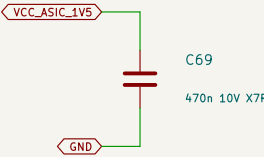
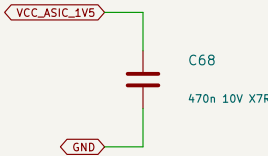
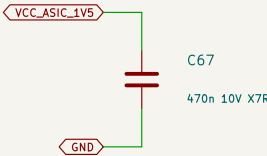
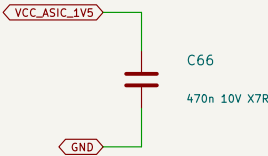
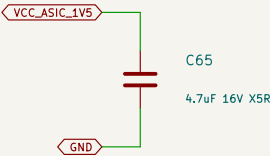
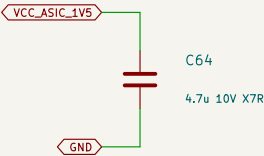
Decoupling / config1.8V and link2.5V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Decoupling / ASIC-facing banks / 1.5 V

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Boot flash

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.

A

B

C

D

E

F

A

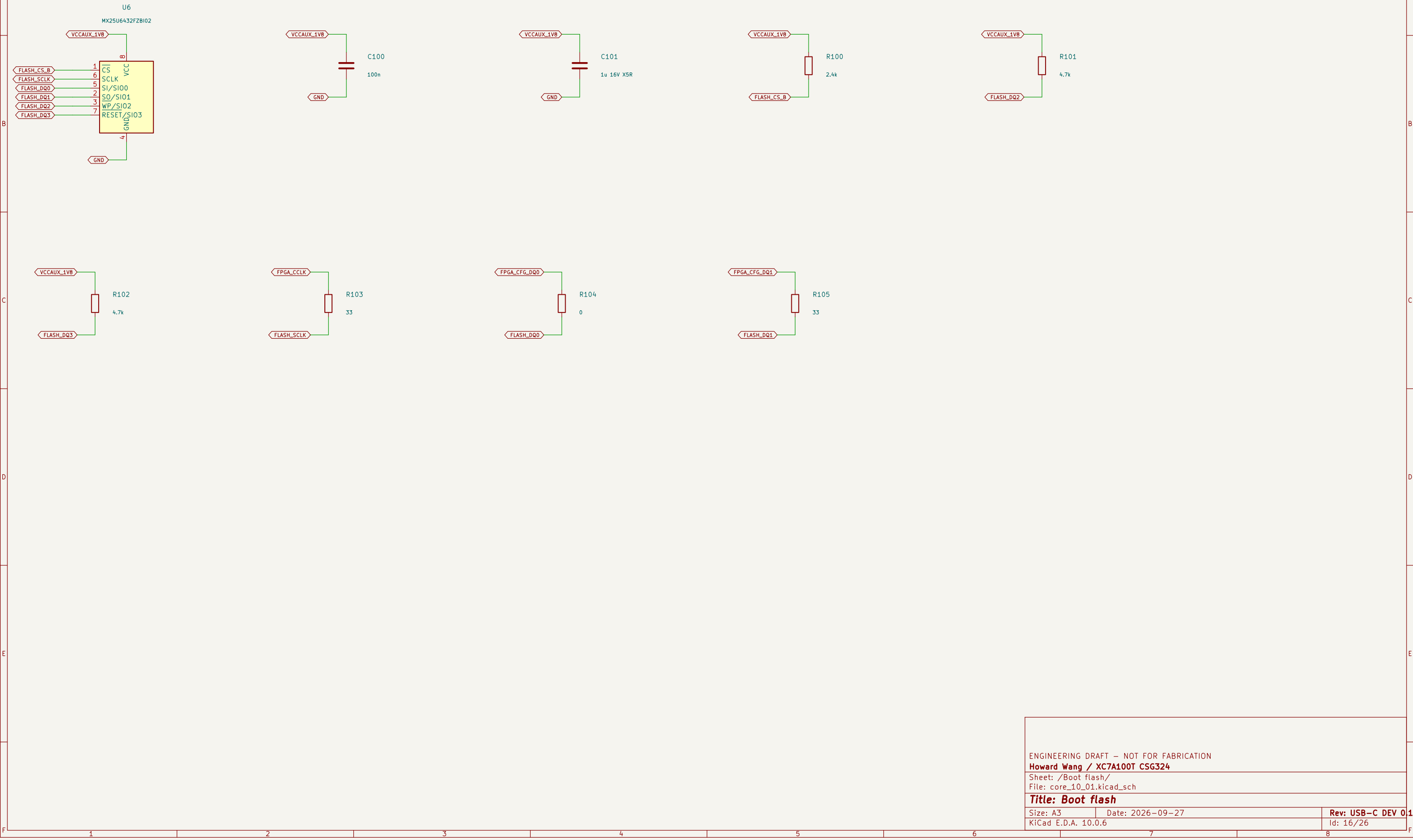
B

C

D

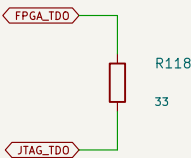
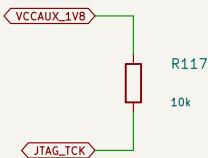
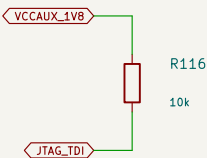
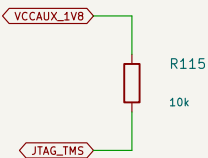
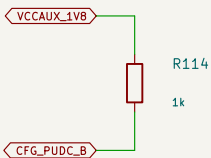
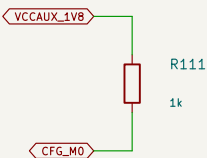
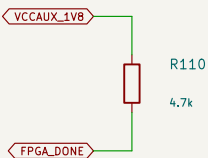
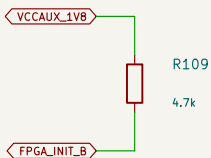
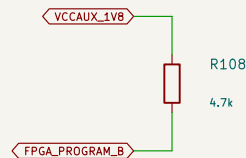
E

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Programming JTAG

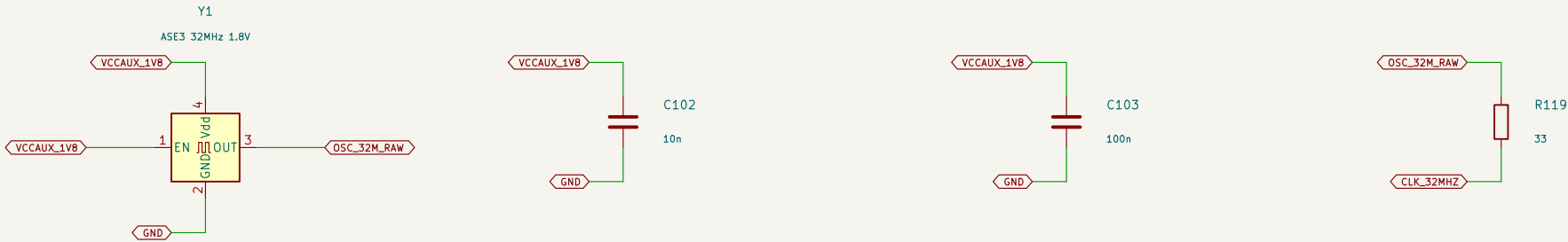
Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



R112 removed: M1/P13 now directly grounded for fixed Master SPI001; JTAG remains available (AMD UG470 v1.17 pp7–8,17,20).

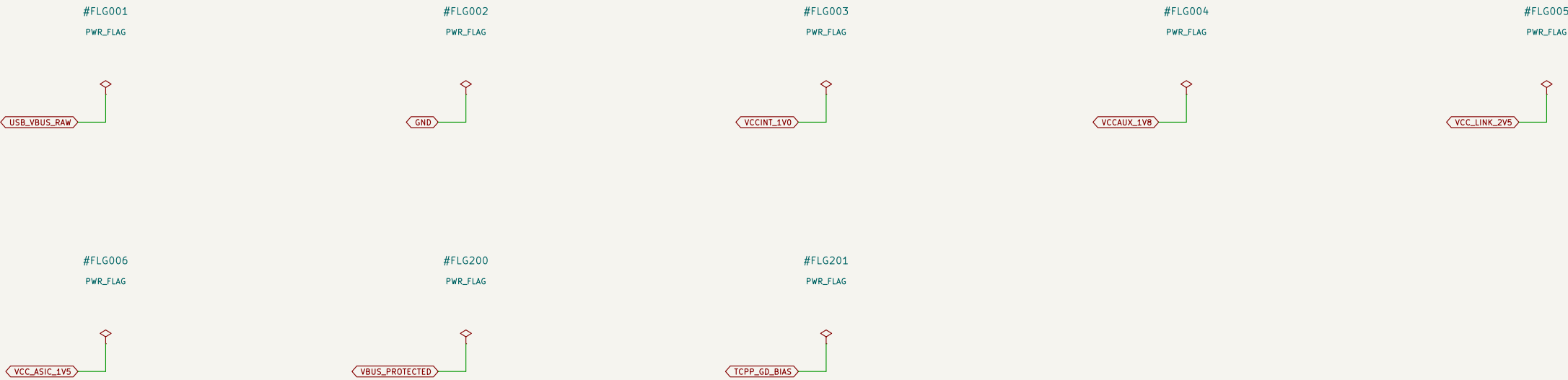
Reference clock

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



Power source assertions

Connected core draft. Matching global labels are electrical connections. Component status fields retain unresolved selections and requirements.



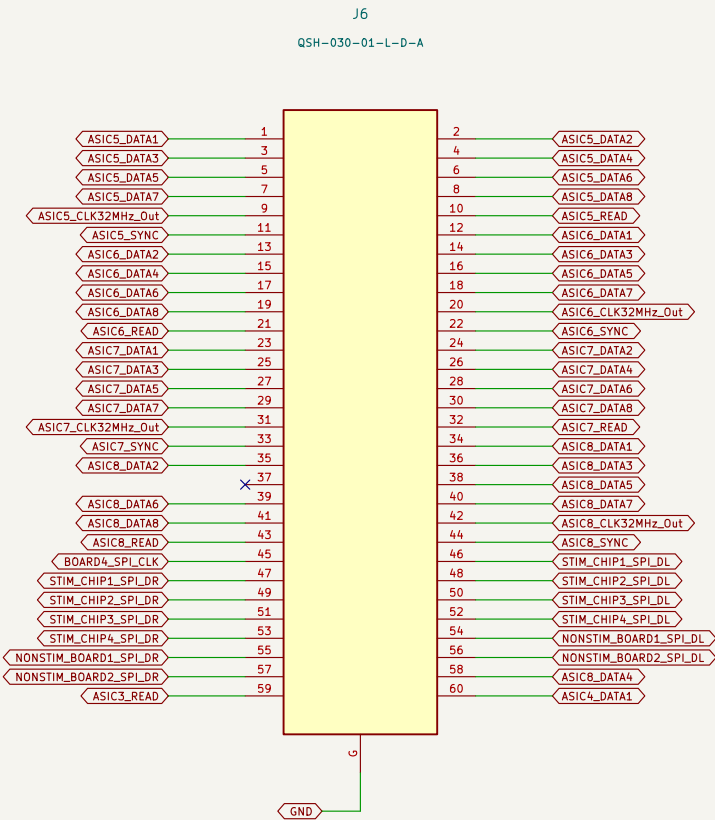
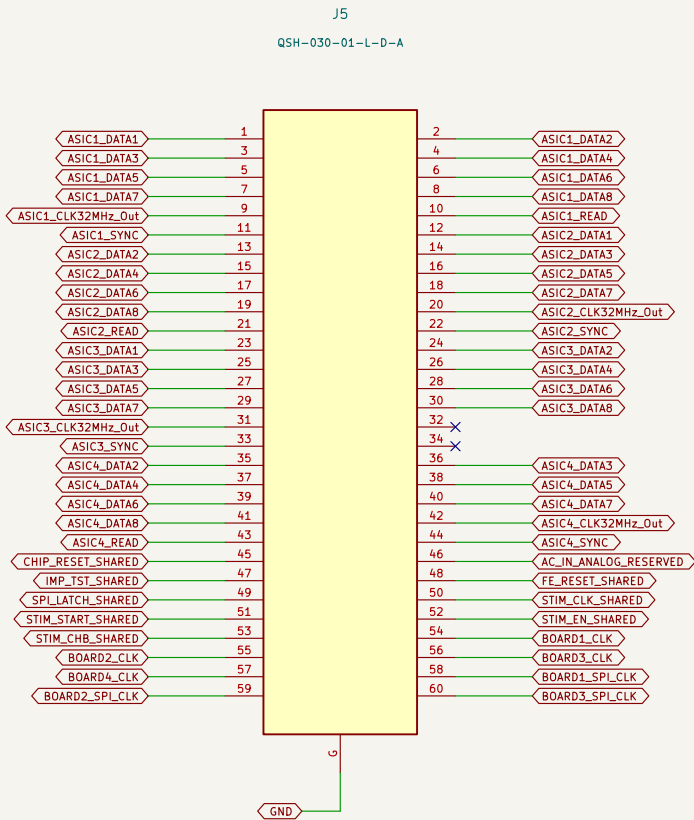
USB-C source flags describe the raw cable supply and conducted FET/diode paths. They do not certify power capacity or startup.

ASIC mezzanine /116 FPGA–linked signals +1 analog contact

117 allocated contacts: 116 digital +1 external analog, J5.32 / J5.34 / J6.37; RESERVED, NOT CONNECTED IN THIS REVISION.

AC_IN analog0–1.5 V: source on routing/ASIC board pending, IMP_TST assumed digital1.5 V; inactive state pending. No ASIC supply on contacts.

J5.46: AC_IN analog0–1.5 V. Routing/ASIC–board analog source pending; no connection to FPGA.



No ASIC rail, analog reference or stimulation supply is allocated on these120 signal contacts.

Electrical thresholds, polarity, safe startup states, physical ASIC Identity and timing remain unapproved. These labels do not certify operation.

USB-C power and protection

USB-C engineering revision. Matching labels are electrical connections. Controller firmware and receiver adapter are required; not a manufacturing release.

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A

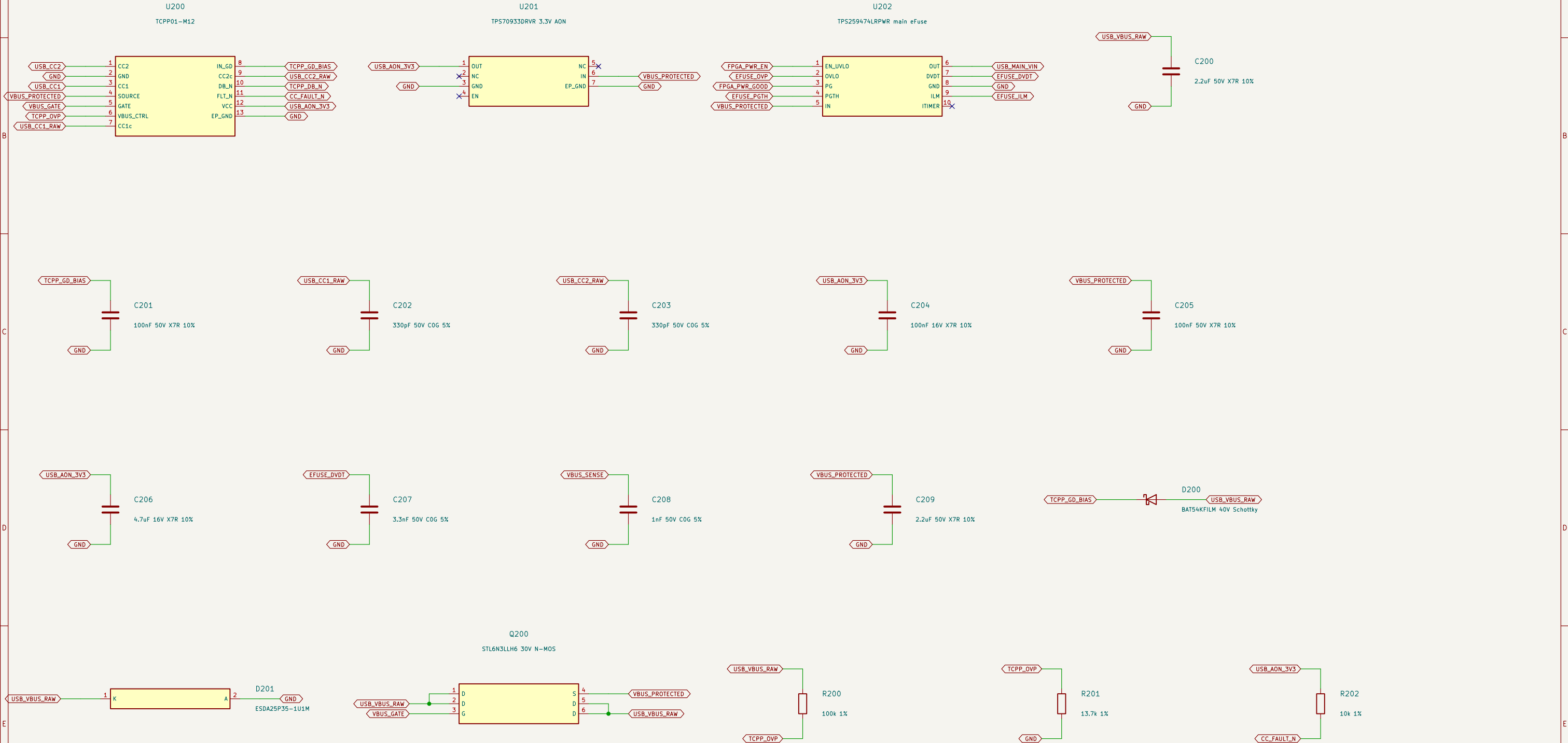
B

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D

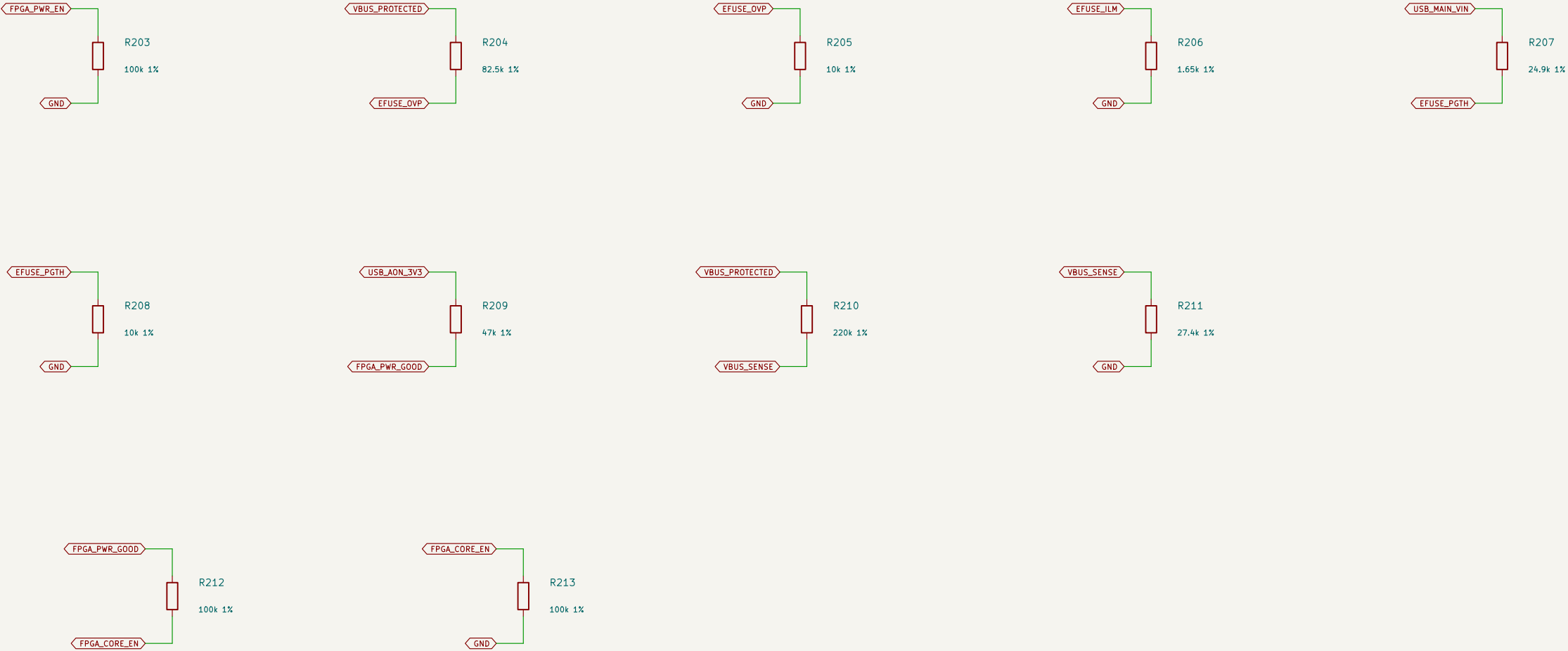
E

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USB-C power and protection / 2

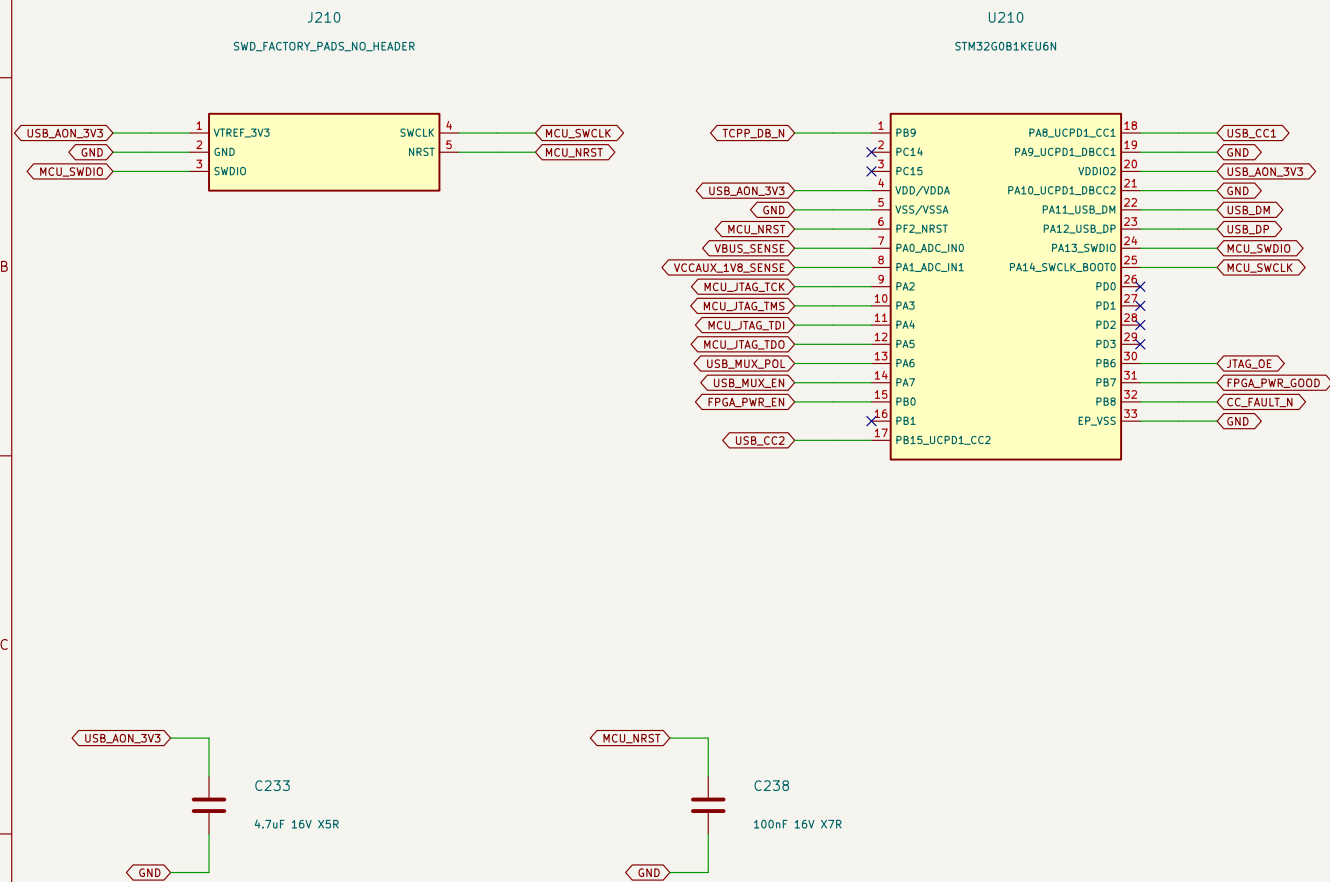
USB-C engineering revision. Matching labels are electrical connections. Controller firmware and receiver adapter are required; not a manufacturing release.



R212/R213 divide the worst-case 1.0 V unpowered PG level below the core EN off threshold. PB7 reads undivided PG.

USB-C controller and recovery fixture

USB-C engineering revision. Matching labels are electrical connections. Controller firmware and receiver adapter are required; not a manufacturing release.



ENGINEERING DRAFT – NOT FOR FABRICATION

Howard Wang / XC7A100T CSG324

Sheet: /USB-C controller and recovery fixture/

File: usb_core_02_01.kicad_sch

Title: USB-C controller and recovery fixture

Size: A3

Date: 2026-09-27

Rev: USB-C DEV 0.1

KiCad E.D.A. 10.0.6

Id: 23/26

USB-C reversible data and ESD

USB-C engineering revision. U213/U214 opposed flow-through lands are unbonded: connect each pair with external copper, never an assumed internal bond.

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A

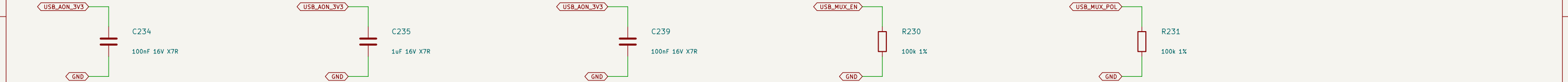
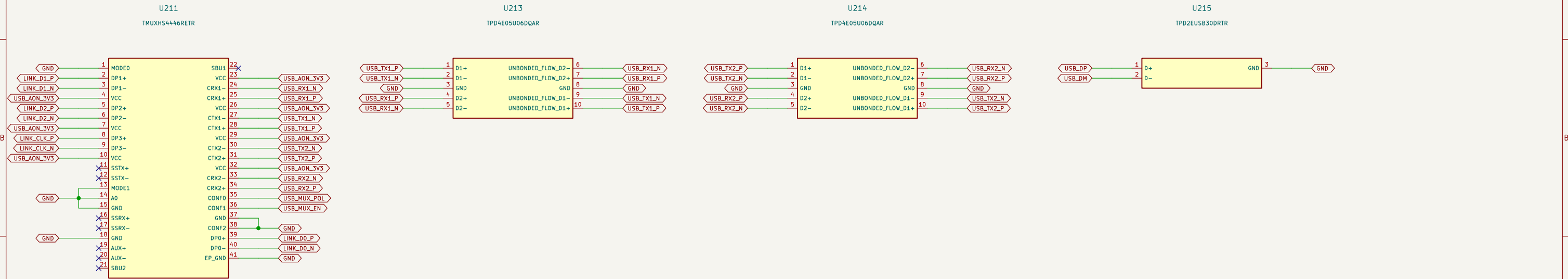
B

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USB-C to 1.8V FPGA JTAG

USB-C engineering revision. Matching labels are electrical connections. Controller firmware and receiver adapter are required; not a manufacturing release.

A

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A

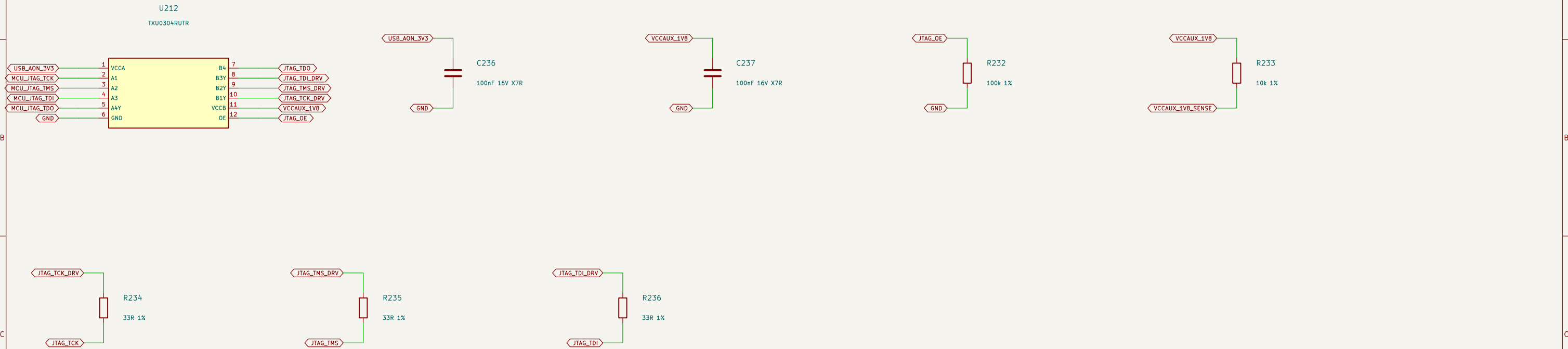
B

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ENGINEERING DRAFT – NOT FOR FABRICATION		
Howard Wang / XC7A100T CSG324		
Sheet: /USB-C to 1.8V FPGA JTAG/ File: usb_core_04_01.kicad_sch		
Title: USB-C to 1.8V FPGA JTAG		
Size: A3	Date: 2026-09-27	Rev: USB-C DEV 0.1
KiCad E.D.A. 10.0.6	Id: 25/26	

USB-C cable connector

USB-C engineering revision. Matching labels are electrical connections. Controller firmware and receiver adapter are required; not a manufacturing release.

A

A

J4

USB-C POWER / DATA / RECOVERY

B

B

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D

E

E

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F

ENGINEERING DRAFT – NOT FOR FABRICATION		
Howard Wang / XC7A100T CSG324		
Sheet: /USB-C cable connector/ File: usb_core_05_01.kicad_sch		
Title: USB-C cable connector		
Size: A3	Date: 2026-09-27	Rev: USB-C DEV 0.1
KiCad E.D.A. 10.0.6		Id: 26/26